

CHAPTER 11

NUMERICAL INTEGRATION

11.1 NUMERICAL INTEGRATION

The need often arises for evaluating the definite integral of a function which has no explicit antiderivative or whose antiderivative is not easily obtained. Moreover, in practical situations, the integrand is often specified in tabular form and then the whole concept of an antiderivative is meaningless.

A definite integral, $\int_a^b f(x) dx$, can be interpreted as the area under the curve $y = f(x)$ bounded by the x axis and the line $x = a$ and $x = b$. In numerical integration we attempt to estimate the area under the curve by evaluating the integrand $f(x)$ at a set of distinct points $\{x_0, x_1, \dots, x_n\}$, where $x_i \in [a, b]$ for $0 \leq i \leq n$.

The numerical integration or computation of a single integral is called mechanical quadrature or quadrature rule, that of double integral, mechanical cubature or cubature rule. The commonly used integration methods can be classified into two groups: the Newton-Cotes formulae that employ functional values at equally spaced points, and the Gaussian quadrature formulae that employ unequally spaced points determined by certain properties of orthogonal polynomials.

To examine the error term involved in quadrature formulae, we introduce the following theorem.

THEOREM 11.1: Let $[c, d]$ contain the points $x_0, x_1, \dots, x_n$ and the interval of integration $[a, b]$. If $f \in C^{n+1}[c, d]$ i.e., $f, f', \dots, f^{(n+1)}$ are continuous on $[c, d]$, then there exists a point $\xi(x) \in [c, d]$ such that

$$E_{n+1}(f) = \frac{1}{(n+1)!} \int_a^b f^{(n+1)}(\xi) F(x) dx$$

where $F(x) = (x-x_0)(x-x_1) \dots (x-x_n)$ and $E_{n+1}(f)$ is the error or remainder term involved in the quadrature formula.

PROOF: By section 4.3, there exists a point $\xi(x) \in [c, d]$ such that

$$f(x) = \sum_{i=0}^n y_i L_i(x) + F(x) \frac{f^{(n+1)}(\xi)}{(n+1)!}$$

where

$$L_i(x) = \frac{(x-x_0)(x-x_1)\dots(x-x_{i-1})(x-x_{i+1})\dots(x-x_n)}{(x_i-x_0)(x_i-x_1)\dots(x_i-x_{i-1})(x_i-x_{i+1})\dots(x_i-x_n)}$$

Integration over $[a, b]$ gives

$$\int_a^b f(x) dx = \sum_{i=0}^n a_i y_i + \frac{1}{(n+1)!} \int_a^b f^{(n+1)}(\xi) F(x) dx$$

where,

$$a_i = \int_a^b L_i(x) dx, \quad i = 0, 1, \dots, n.$$

Thus

$$\begin{aligned} E_{n+1}(f) &= \int_a^b f(x) dx - \sum_{i=0}^n a_i y_i \\ &= \frac{1}{(n+1)!} \int_a^b f^{(n+1)}(\xi) F(x) dx \end{aligned} \quad (11.1)$$

If we denote $M_{n+1} = \max_{a \leq x \leq b} |f^{(n+1)}(x)|$

a bound is obtained of the form

$$|E_{n+1}(f)| \leq \frac{M_{n+1}}{(n+1)!} \int_a^b |F(x)| dx.$$

THEOREM 11.2: Let the interval $[c,d]$ contain the points $x_0, x_1, \dots, x_n$ and the interval $[a,b]$. If $f \in C^{n+1}[c,d]$ and the product $(x-x_0)(x-x_1)\dots(x-x_n)$ does not change sign on (a,b) , then there is a point $\mu \in [a,b]$ such that

$$E_{n+1}(f) = I(f) - \sum_{i=0}^n a_i y_i = \frac{f^{(n+1)}(\mu)}{(n+1)!} \int_a^b F(x) dx. \quad (11.2)$$

PROOF: From Eq. (11.1),

$$E_{n+1}(f) = \frac{1}{(n+1)!} \int_a^b f^{(n+1)}(\xi) F(x) dx.$$

Since $f^{(n+1)}(\xi)$ is a continuous function of x and $F(x)$ does not change sign, by the Mean Value Theorem for integrals (Appendix A) there exists a point $\mu \in [a,b]$ such that

$$\int_a^b f^{(n+1)}(\xi) F(x) dx = f^{(n+1)}(\mu) \int_a^b F(x) dx$$

and hence

$$E_{n+1}(f) = \frac{1}{(n+1)!} f^{(n+1)}(\mu) \int_a^b F(x) dx.$$

The method of quadrature will be based on interpolation polynomials (chapters 4 & 5) appropriate for a given function.

Suppose, for a function $y = f(x)$, we know the corresponding values at $n+1$ points $x_0, x_1, \dots, x_n$ of $[a,b]$:

$$y_i = f(x_i) \quad \text{for } i = 0, 1, 2, \dots, n.$$

It is required to find approximately

$$I(x) = \int_a^b y \, dx = \int_a^b f(x) \, dx.$$

Let $P_n(x)$ be the Lagrange interpolating polynomial

$$P_n(x) = \sum_{l=0}^n f(x_l) L_l(x)$$

where

$$L_l(x) = \frac{(x-x_0)(x-x_1)\dots(x-x_{l-1})(x-x_{l+1})\dots(x-x_n)}{(x_l-x_0)(x_l-x_1)\dots(x_l-x_{l-1})(x_l-x_{l+1})\dots(x_l-x_n)}$$

and $P_n(x_l) = f(x_l)$, $l = 0, 1, \dots, n$.

Replacing the function $f(x)$ by the polynomial $P_n(x)$, we obtain

$$\int_a^b f(x) dx = \int_a^b P_n(x) dx + E_{n+1}(f)$$

or

$$\begin{aligned} I(f) &= \int_a^b \sum_{l=0}^n f(x_l) L_l(x) dx + E_{n+1}(f) \\ &= \sum_{l=0}^n \left[\int_a^b L_l(x) dx \right] f(x_l) + E_{n+1}(f) \end{aligned}$$

or

$$I(f) = \sum_{l=0}^n a_l f(x_l) + E_{n+1}(f) \quad (11.3)$$

where $E_{n+1}(f)$ is the error or remainder term in the quadrature formula (11.3) and

$$a_l = \int_a^b L_l(x) dx, \quad l = 0, 1, 2, \dots, n.$$

From the formula (11.3), we get the approximate quadrature formula

$$\int_a^b f(x) dx \approx \sum_{l=0}^n a_l f(x_l) \quad (11.4)$$

where

$$a_l = \int_a^b L_l(x) dx, \quad l = 0, 1, 2, \dots, n.$$

If the integration limits a and b are interpolating points,

then the quadrature formulae are of the "closed type", otherwise they are the "open type".

11.2 THE TRAPEZOIDAL RULE WITH ERROR TERM

One of the simplest ways to estimate an integral $I = \int_a^b f(x)dx$ is to employ linear interpolation, i.e., to approximate the curve $y = f(x)$ by a straight line $y = P_1(x)$ passing through the points $(a, f(a))$ and $(b, f(b))$ and then to compute the area under the line (see Fig. 11.1).

Let $x_0 = a$, $x_1 = b$, and $h = b - a$. To approximate

$$\int_a^b f(x)dx = \int_{x_0}^{x_1} f(x)dx,$$

we use the formula (11.4) for $n = 1$

$$\begin{aligned} \int_{x_0}^{x_1} P_1(x)dx &= \int_{x_0}^{x_1} \left[\frac{x - x_1}{x_0 - x_1} f(x_0) + \frac{x - x_0}{x_1 - x_0} f(x_1) \right] dx \\ &= \frac{-f(x_0)}{h} \int_{x_0}^{x_1} (x - x_1)dx + \frac{f(x_1)}{h} \int_{x_0}^{x_1} (x - x_0)dx. \end{aligned}$$

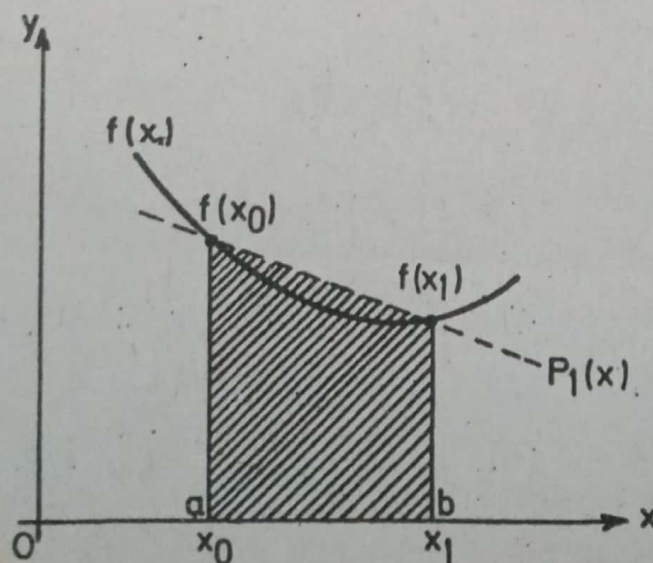


Fig. 11.1 The Trapezoidal Rule.

Using the change of variable $x = x_0 + th$, we obtain

$$\int_{x_0}^{x_1} p_1(x) dx = -hf(x_0) \int_0^1 (t-1) dt + hf(x_1) \int_0^1 t dt = \frac{h}{2} [f(x_0) + f(x_1)]$$

The quadrature formula

$$\int_{x_0}^{x_1} f(x) dx \approx \frac{h}{2} [f(x_0) + f(x_1)] \quad (11.5)$$

is called the **trapezoidal rule** or **trapezoidal formula**.

Let us calculate the error term involved in the quadrature formula (11.5).

Since $(x-x_0)(x-x_1)$ is always negative on (x_0, x_1) , Theorem 11.2 implies that, for $f \in C^2[x_0, x_1]$, there is a $\xi \in [x_0, x_1]$ such that

$$E_2(f) = \frac{f''(\xi)}{2!} \int_{x_0}^{x_1} (x - x_0)(x - x_1) dx.$$

After using the change of variable $x = x_0 + ht$ and integrating, we have

$$E_2(f) = -\frac{h^3}{12} f''(\xi).$$

Thus the trapezoidal rule with error term is

$$\int_{x_0}^{x_1} f(x) dx = \frac{h}{2} [f(x_0) + f(x_1)] - \frac{h^3}{12} f''(\xi). \quad (11.6)$$

The trapezoidal rule is exact if $f(x)$ is linear, since in this situation the second and higher order derivatives are zero.

EXAMPLE 11.1: Evaluate $I = \int_0^1 (x + 1)dx$

using the trapezoidal rule.

SOLUTION: $I = \int_0^1 (x + 1)dx = 3/2$

By the trapezoidal rule,

$$\int_0^1 (x + 1)dx = \frac{(1-0)}{2} [f(0) + f(1)] = \frac{1}{2}(1 + 2) = \frac{3}{2}.$$

Hence trapezoidal rule is exact when $f(x)$ is linear.

EXAMPLE 11.2: Evaluate $I = \int_0^1 3x^2 dx$

using the trapezoidal rule. Compare with the correct value of the integral.

SOLUTION: $I = \int_0^1 3x^2 dx = 1.$

By the trapezoidal rule, we have

$$\int_0^1 3x^2 dx \approx \frac{(1-0)}{2} [f(0) + f(1)] = \frac{1}{2} (0 + 3) = 1.5.$$

Thus

$$\text{Actual error} = 1 - 1.5 = -0.5.$$

The following theorem is stated (without proof) for the quadrature formulae with equally spaced points for the improvement of error.

THEOREM 11.3: Let $x_0 = a$, $x_n = b$, and $h = \frac{b-a}{n}$. There exists a point $\xi \in [a, b]$ such that

$$\int_a^b f(x)dx = \sum_{i=0}^n a_i f(x_i) + \frac{h^{n+2}}{(n+1)!} f^{(n+1)}(\xi) \int_0^n t(t-1)\dots(t-n)dt \quad (11.7)$$

if n is odd and $f \in C^{n+1}[a, b]$, and

$$\int_a^b f(x)dx = \sum_{i=0}^n a_i f(x_i) + \frac{h^{n+3}}{(n+2)!} f^{(n+2)}(\xi) \int_0^n t^2(t-1)\dots(t-n)dt \quad (11.8)$$

if n is even and $f \in C^{n+2}[a, b]$.

In Theorem 11.3, note that, when n is an even integer, the degree of precision is $(n+1)$ although the interpolation polynomial is of degree at most n . When n is odd, the degree of precision is only n . Consequently, if n is even and more points are to be added to increase precision, no accuracy is gained by adding only one point and points should be added in multiples of two.

11.3 SIMPSON'S 1/3 RULE WITH ERROR TERM

The trapezoidal rule tries to simplify integration by approximating the function to be integrated by a straight line or a series of straight line segments. In Simpson's rule we try to approximate by a series of parabolic segments, hoping that the parabola will more closely match a given curve $f(x)$, than would the straight line in the trapezoidal rule. To estimate $I = \int_a^b f(x)$, the curve $y=f(x)$, is approximated by a parabola $y = P_2(x)$ passing through three points $(x_0, f(x_0))$, $(x_1, f(x_1))$, $(x_2, f(x_2))$ and then the area under the parabolic segments is computed (see Fig.11.2).

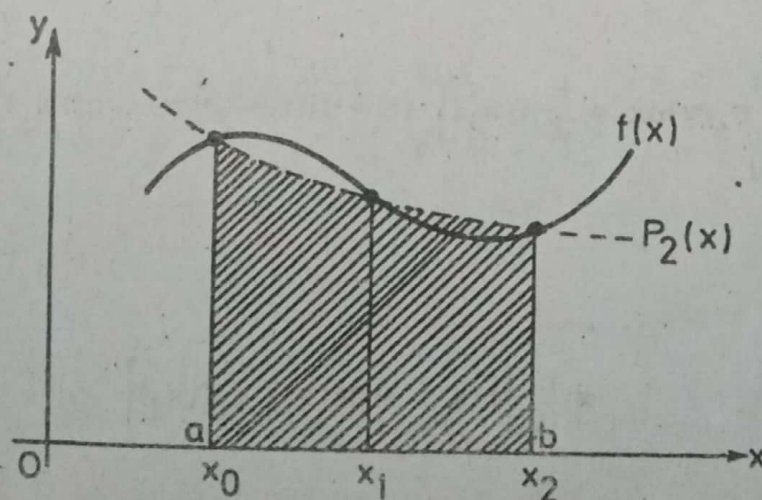


Fig.11.2 Simpson's Rule

Let $x_0 = a$, $h = \frac{b-a}{2}$ so that $x_1 = x_0 + h$, $x_2 = x_0 + 2h = b$.

Then

$$\int_a^b f(x)dx = \int_{x_0}^{x_2} f(x)dx.$$

Applying formula (11.4), we obtain for $n = 2$,

$$\begin{aligned} \int_{x_0}^{x_2} P_2(x)dx &= \int_{x_0}^{x_2} \left[\frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} f(x_0) + \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} f(x_1) \right. \\ &\quad \left. + \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} f(x_2) \right] dx \\ &= \frac{f(x_0)}{2h^2} \int_{x_0}^{x_2} (x-x_1)(x-x_2)dx - \frac{f(x_1)}{h^2} \int_{x_0}^{x_2} (x-x_0)(x-x_2)dx \\ &\quad + \frac{f(x_2)}{2h^2} \int_{x_0}^{x_2} (x-x_0)(x-x_1)dx. \end{aligned}$$

Substituting $x = x_0 + th$, then $dx = h dt$, we get

$$\begin{aligned} \int_{x_0}^{x_2} P_2(x)dx &= \frac{h}{2} f(x_0) \int_0^2 (t-1)(t-2)dt - hf(x_1) \int_0^2 t(t-2)dt \\ &\quad + \frac{h}{2} f(x_2) \int_0^2 t(t-1)dt \\ &= \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)] \end{aligned}$$

This quadrature formula

Let $x_0 = a$, $h = \frac{b-a}{2}$ so that $x_1 = x_0 + h$, $x_2 = x_0 + 2h = b$.

Then

$$\int_a^b f(x)dx = \int_{x_0}^{x_2} f(x)dx.$$

Applying formula (11.4), we obtain for $n = 2$,

$$\begin{aligned} \int_{x_0}^{x_2} P_2(x)dx &= \int_{x_0}^{x_2} \left[\frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} f(x_0) + \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} f(x_1) \right. \\ &\quad \left. + \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} f(x_2) \right] dx \\ &= \frac{f(x_0)}{2h^2} \int_{x_0}^{x_2} (x-x_1)(x-x_2)dx - \frac{f(x_1)}{h^2} \int_{x_0}^{x_2} (x-x_0)(x-x_2)dx \\ &\quad + \frac{f(x_2)}{2h^2} \int_{x_0}^{x_2} (x-x_0)(x-x_1)dx. \end{aligned}$$

Substituting $x = x_0 + th$, then $dx = h dt$, we get

$$\begin{aligned} \int_{x_0}^{x_2} P_2(x)dx &= \frac{h}{2} f(x_0) \int_0^2 (t-1)(t-2)dt - hf(x_1) \int_0^2 t(t-2)dt \\ &\quad + \frac{h}{2} f(x_2) \int_0^2 t(t-1)dt \\ &= \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)] \end{aligned}$$

This quadrature formula

$$\int_{x_0}^{x_2} f(x) dx \approx \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)] \quad (11.9)$$

is called Simpson's 1/3 rule.

If $f \in C^{(4)}[x_0, x_2]$, the Theorem 11.3 (Eq. 11.8) implies that there exists a point $\xi \in [x_0, x_2]$ such that

$$\begin{aligned} E_3(f) &= \frac{h^5 f^{(4)}(\xi)}{4!} \int_0^2 t^2(t-1)(t-2) dt \\ &= -\frac{h^5 f^{(4)}(\xi)}{90}. \end{aligned}$$

Thus, Simpson's 1/3 rule with error term is

$$\int_{x_0}^{x_2} f(x) dx = \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)] - \frac{h^5 f^{(4)}(\xi)}{90}. \quad (11.10)$$

11.4 SIMPSON'S 3/8 RULE WITH ERROR TERM

Let $x_0 = a$, $h = \frac{b-a}{3}$ so that $x_1 = x_0 + h$, $x_2 = x_0 + 2h$, $x_3 = x_0 + 3h = b$.

We have

$$\int_a^b f(x) dx = \int_{x_0}^{x_3} f(x) dx.$$

For $n = 3$, we get from the formula (11.4)

$$\begin{aligned} \int_{x_0}^{x_3} P_3(x) dx &= \int_{x_0}^{x_3} \left[\frac{(x - x_1)(x - x_2)(x - x_3)}{(x_0 - x_1)(x_0 - x_2)(x_0 - x_3)} f(x_0) + \right. \\ &\quad \left. \frac{(x - x_0)(x - x_2)(x - x_3)}{(x_1 - x_0)(x_1 - x_2)(x_1 - x_3)} f(x_1) + \right. \\ &\quad \left. \frac{(x - x_0)(x - x_1)(x - x_3)}{(x_2 - x_0)(x_2 - x_1)(x_2 - x_3)} f(x_2) + \right. \\ &\quad \left. \frac{(x - x_0)(x - x_1)(x - x_2)}{(x_3 - x_0)(x_3 - x_1)(x_3 - x_2)} f(x_3) \right] dx \end{aligned}$$

$$\frac{(x - x_0)(x - x_1)(x - x_3)}{(x_2 - x_0)(x_2 - x_1)(x_2 - x_3)} f(x_2) + \frac{(x - x_0)(x - x_1)(x - x_2)}{(x_3 - x_0)(x_3 - x_1)(x_3 - x_2)} f(x_3) \Big] dx$$

$$\begin{aligned} \int_{x_0}^{x_3} P_3(x) dx &= - \frac{f(x_0)}{6h^3} \int_{x_0}^{x_3} (x - x_1)(x - x_2)(x - x_3) dx + \\ &\quad \frac{f(x_1)}{2h^3} \int_{x_0}^{x_3} (x - x_0)(x - x_2)(x - x_3) dx - \\ &\quad \frac{f(x_2)}{2h^3} \int_{x_0}^{x_3} (x - x_0)(x - x_1)(x - x_3) dx + \\ &\quad \frac{f(x_3)}{6h^3} \int_{x_0}^{x_3} (x - x_0)(x - x_1)(x - x_2) dx. \end{aligned}$$

Using the change of variable $x = x_0 + ht$, we get

$$\begin{aligned} \int_{x_0}^{x_3} P_3(x) dx &= \frac{-hf(x_0)}{6} \int_0^3 (t-1)(t-2)(t-3) dt + \frac{hf(x_1)}{2} \int_0^3 t(t-2)(t-3) dt \\ &\quad - \frac{hf(x_2)}{2} \int_0^3 t(t-1)(t-3) dt + \frac{hf(x_3)}{6} \int_0^3 t(t-1)(t-2) dt \\ &= \frac{hf(x_0)}{6} \frac{9}{4} + \frac{hf(x_1)}{2} \frac{9}{4} - \frac{hf(x_2)}{2} \left(-\frac{9}{4}\right) + \frac{hf(x_3)}{6} \frac{9}{4} \\ &= \frac{3}{8} h \left[f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3) \right]. \end{aligned}$$

The integration formula

$$\int_{x_0}^{x_3} f(x) dx \approx \frac{3}{8} h [f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)] \quad (11.11)$$

s called Simpson's 3/8 rule.

The Theorem 11.3 (Eq. 11.7) implies that, for $f \in C^4[x_0, x_3]$, there exists a point $\xi \in [x_0, x_3]$ such that

$$\begin{aligned} E_4(f) &= \frac{h^5 f^{(4)}(\xi)}{4!} \int_0^3 t(t-1)(t-2)(t-3) dt \\ &= \frac{h^5 f^{(4)}(\xi)}{24} \int_0^3 (t^4 - 6t^3 + 11t^2 - 6t) dt \\ &= -\frac{3}{80} h^5 f^{(4)}(\xi). \end{aligned}$$

Thus

$$\int_{x_0}^{x_3} f(x) dx = \frac{3}{8} h [f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)] - \frac{3}{80} h^5 f^{(4)}(\xi) \quad (11.12)$$

s called Simpson's 3/8 rule with error term.

The rules (11.9) and (11.11) are exact for all polynomials of degree three or less as their error terms involve 4th derivative of $f(x)$ which vanish identically.

EXAMPLE 11.3: Use the trapezoidal and Simpson's rules to estimate the integral

$$\int_1^3 f(x) dx = \int_1^3 (x^3 - 2x^2 + 7x - 5) dx.$$

SOLUTION:
$$\int_1^3 f(x) dx = \int_1^3 (x^3 - 2x^2 + 7x - 5) dx$$

$$= \left[\frac{x^4}{4} - \frac{2x^3}{3} + \frac{7x^2}{2} - 5x \right]_1^3 = 20 \frac{2}{3}.$$

For the trapezoidal rule (11.6), $x_0 = 1$, $x_1 = 3$ so that

$$h = 3 - 1 = 2,$$

$$\begin{aligned}\int_1^3 f(x) dx &= \frac{h}{2} [f(1) + f(3)] - \frac{h^3}{12} f''(\xi) \\ &= \frac{2}{2} [1 + 25] - \frac{2}{3} f''(\xi) = 26 - \frac{2}{3} f''(\xi).\end{aligned}$$

Since $f''(x) = 6x - 4$, the error is given by

$$-\frac{2}{3} f''(\xi) = -\frac{2}{3} (6\xi - 4), \quad \xi \in [1, 3]$$

which has an extreme value $-9 \frac{1}{3}$ for $\xi = 3$.

The actual error $= 20\frac{2}{3} - 26 = -5\frac{1}{3}$, which is smaller than the bound as expected, but still quite large.

For Simpson's $\frac{1}{3}$ rule (11.10), $x_0 = 1$, $h = \frac{3-1}{2} = 1$, $x_1 = 2$, $x_2 = 3$,

$$\begin{aligned}\int_1^3 f(x) dx &= \frac{h}{3} [f(1) + 4f(2) + f(3)] - \frac{h^5}{90} f^{(4)}(\xi) \\ &= \frac{1}{3} [1 + 4 \times 9 + 25] - \frac{1}{90} f^{(4)}(\xi) = 20\frac{2}{3} - \frac{1}{90} f^{(4)}(\xi).\end{aligned}$$

Since the fourth derivative of $f(x)$ vanishes for all x , the error term vanishes and the result is exact.

For Simpson's $\frac{3}{8}$ rule, $x_0 = 1$, $h = \frac{3-1}{3} = \frac{2}{3}$, $x_1 = \frac{5}{3}$, $x_2 = \frac{7}{3}$, $x_3 = 3$,

$$\begin{aligned}\int_1^3 f(x) dx &= \frac{3}{8} h \left[f(1) + 3f\left(\frac{5}{3}\right) + 3f\left(\frac{7}{3}\right) + f(3) \right] - \frac{3}{80} h^5 f^{(4)}(\xi) \\ &= \frac{3}{8} \times \frac{2}{3} \left[1 + 3 \times \frac{155}{27} + 3 \times \frac{355}{27} + 25 \right] - \frac{3}{80} \left(\frac{2}{3}\right)^5 f^{(4)}(\xi) \\ &= \frac{1}{4} \left[\frac{9 + 155 + 355 + 225}{9} \right] - \frac{3}{80} \left(\frac{2}{3}\right)^5 f^{(4)}(\xi) \\ &= \frac{1}{4} \left(\frac{744}{9}\right) - \frac{3}{80} \left(\frac{2}{3}\right)^5 f^{(4)}(\xi) = 20\frac{2}{3} - \frac{3}{80} \left(\frac{2}{3}\right)^5 f^{(4)}(\xi).\end{aligned}$$

The result is exact again as the 4th derivative of $f(x)$ vanishes identically.

11.5 ERROR ESTIMATION

Before proceeding further we will consider an easy way to estimate the error involved in the quadrature formulae.

The trapezoid rule (11.5), for example, is exact if $f(x)$ is linear. If this is not the case, however, there will be an error which is about $-\frac{h^3}{12} f''(\xi)$. The value of ξ at which $f''(\xi)$ should be evaluated is not known, except that it lies within the range of integration. This expression can thus only be used to determine an upper bound for the error and then only if we have an exact or approximate expression for $f(x)$. We do learn, however, that the error is approximately proportional to h^3 .

If we confine ourselves to an examination of quadrature formulae with equally spaced points, the accuracy of the quadrature formula, in this case, is mainly characterized by the order of the error term

$$E_{n+1}(f) = O(h^m) \quad (11.13)$$

where

$$h = \frac{b - a}{n}$$

is the spacing (n is the number of divisions) and m is a natural number. For example, for the trapezoidal rule (Sec. 11.2) we have

$$E_2(f) = -\frac{h^3}{12} f''(\xi) = -\frac{b-a}{12} h^2 f''(\xi)$$

and so $m = 2$ giving the relation

$$E_2(f) = O(h^2),$$

that is, the trapezoidal rule is a 2nd order approximation process. For Simpson's rule (Sec. 11.3) we have

$$E_3(f) = -\frac{b-a}{180} h^4 f^{(4)}(\xi)$$

and hence $m = 4$. Thus $E_3(f) = O(h^4)$,

that is, the Simpson's rule is a fourth order process.

EXAMPLE 11.4: Find the upper bound for error term in the trapezoidal rule for the integral

$$I = \int_1^2 \left(\frac{e^{-x}}{x} \right) dx.$$

SOLUTION: We have

$$E_2(f) = -\frac{h^3}{12} f''(\xi).$$

Also

$$f(x) = \frac{e^{-x}}{x}, \quad f'(x) = -\left(\frac{1}{x} + \frac{1}{x^2}\right)e^{-x}, \quad f''(x) = \left(\frac{1}{x} + \frac{2}{x^2} + \frac{2}{x^3}\right)e^{-x}.$$

The second derivative attains its maximum value on $[1, 2]$ at $x = 1$; hence we can take

$$M_2 = \max_{x \in [1, 2]} f''(x) = f''(1) = 5e^{-1} \approx 1.84.$$

$$\text{Thus error bound} = \frac{h^3}{12} M_2 = \frac{1}{12} (1.84) = 0.1533.$$

Simpson's rule
Trapezoidal rule
 $h = \frac{b-a}{n}$
 $3.14 - 0$

EXERCISE 11.1

$n = 4$

1. The integral $\int_0^\pi \sin x \, dx$ is evaluated by interpolating the function $f(x) = \sin x$ at the points $x = 0$ and $x = \pi$. Calculate the bound for the error and compare it with the actual error.

2. Use the trapezoidal and Simpson's 1/3 rules to approximate

$$\int_0^{\pi/2} \sin^2 x \, dx. \text{ Find a maximum bound for the error in each case.}$$

Compare your approximations with the actual result.

3. Approximate $\int_0^1 x^{1/3} dx$, using the trapezoidal and Simpson's 1/3 rules. Compare the approximations with the actual value.
4. Approximate $\int_1^2 \ln x dx$ using the trapezoidal and Simpson's rules. Compare the approximations with the actual value.
5. Use the table below to find an approximation to

$$\int_{1.1}^{1.5} e^x dx, \text{ using:}$$

- a) The trapezoidal rule with $x_0 = 1.1$, and $x_1 = 1.5$:
 b) Simpson's rule with $x_0 = 1.1$, $x_1 = 1.3$, and $x_2 = 1.5$.

x	1.1	1.3	1.5
e^x	3.0042	3.6693	4.4817

6. Show that the trapezoidal rule yields for $n > 1$ the exact value of the integrals

$$\int_0^{2\pi} \cos x dx, \int_0^{2\pi} \sin x dx.$$

7. Consider the computation of $I = \int_a^b f(x) dx$, and suppose that I_1 and I_2 denote the estimates of I obtained using Simpson's 1/3 and 3/8 rules, respectively, with step sizes $h_1 = (b - a)/2$ and $h_2 = (b - a)/3$, respectively.

a) Show that (subject to appropriate assumptions concerning the truncation errors) a better estimate of I is given by

$$I^* \approx \frac{9}{5} I_2 - \frac{4}{5} I_1.$$

- b) If $f(x) = x^{-2}$, $a=1$ and $b=3$, find the values of I_1, I_2 and I^* . What are the percentage errors in the three values?

8. Approximate the integral $\int_1^2 \frac{1}{x} dx$, using the trapezoidal rule and Simpson's rule.

11.6 COMPOSITE NUMERICAL INTEGRATION

One of the ways to reduce the error associated with a low order integration formula is to subdivide the interval of integration $[a,b]$ into smaller intervals and then to use the formula separately on each sub interval. Repeated application of a low-order formula is usually preferred to the single application of a high order formula, partly because of the simplicity of the low-order formulae and partly because of computational difficulties associated with the high-order formulae. For these reasons it is generally inappropriate to use a single polynomial of high degree (high-order formula) to approximate an integral over a large interval.

Consider finding an approximation to $\int_0^4 e^x dx$. The interval $[0,4]$ would be considered a fairly large interval. A high-order formula is likely to give an inaccurate approximation to this integral, or would be difficult to evaluate. Using a low-order formula, however, will not solve the problem.

Now

$$\int_0^4 e^x dx = e^4 - e^0 = 53.59815.$$

Suppose Simpson's rule is used with $n = 2$ so that $h = \frac{4-0}{2} = 2$.

Then

$$\int_0^4 e^x dx \approx \frac{2}{3} [e^0 + 4e^2 + e^4] = 56.76958.$$

The actual error = $53.59815 - 56.76958 = -3.17143$ is far larger than would generally be regarded as acceptable.

Divide the interval $[0,4]$ into subintervals $[0,2]$ and $[2,4]$ and use Simpson's rule on each subinterval with $h = 1$. Thus

$$\begin{aligned} \int_0^4 e^x dx &= \int_0^2 e^x dx + \int_2^4 e^x dx \\ &\approx \frac{1}{3} [e^0 + 4e + e^2] + \frac{1}{3} [e^2 + 4e^3 + e^4] \\ &= \frac{1}{3} [e^0 + 4e + 2e^2 + 4e^3 + e^4] = 53.86385. \end{aligned}$$

The actual error has been reduced to -0.26570 . We subdivide each of the above subintervals and use Simpson's rule with $h = 1/2$, giving

$$\begin{aligned}\int_0^4 e^x dx &= \int_0^1 e^x dx + \int_1^2 e^x dx + \int_2^3 e^x dx + \int_3^4 e^x dx \\ &\approx \frac{1}{6} [e^0 + 4e^{1/2} + e] + \frac{1}{6} [e + 4e^{3/2} + e^2] + \\ &\quad \frac{1}{6} [e^2 + 4e^{5/2} + e^3] + \frac{1}{6} [e^3 + 4e^{7/2} + e^4] \\ &= \frac{1}{6} [e^0 + 4e^{1/2} + 2e + 4e^{3/2} + 2e^2 + 4e^{5/2} + 2e^3 + 4e^{7/2} + e^4] \\ &= 53.61622.\end{aligned}$$

The actual error for this approximation is -0.01807 .

Integration formulae resulting from interval subdivision and repeated application of low-order formula are called **composite integration formulae**.

11.6.1 COMPOSITE TRAPEZOIDAL RULE

Let $f \in C^2[a, b]$. Subdivide the interval of integration $[a, b]$ into n equal subintervals $[x_0, x_1], [x_1, x_2], \dots, [x_{n-1}, x_n]$ and then use the trapezoidal rule on each subinterval (see Fig. 11.3). With $h = \frac{b-a}{n}$ and $x_i = a + ih$, $i = 0, 1, \dots, n$,

$$\begin{aligned}\int_a^b f(x) dx &= \int_{x_0}^{x_n} f(x) dx \\ &= \int_{x_0}^{x_1} f(x) dx + \int_{x_1}^{x_2} f(x) dx + \dots + \int_{x_{n-1}}^{x_n} f(x) dx \\ &= \frac{h}{2} [f(x_0) + f(x_1)] + \frac{h}{2} [f(x_1) + f(x_2)] + \dots + \frac{h}{2} [f(x_{n-1}) + f(x_n)] \\ &\quad - \frac{h^3}{12} \sum_{i=1}^n f''(\bar{\xi}_i), \quad \bar{\xi}_i \in (x_{i-1}, x_i)\end{aligned}$$

$$\int_a^b f(x) dx = \frac{h}{2} \left[f(x_0) + f(x_n) + 2 \sum_{i=1}^{n-1} f(x_i) \right] - \frac{h^3}{12} \sum_{i=1}^n f''(\bar{\xi}_i), \quad \bar{\xi}_i \in (x_{i-1}, x_i)$$

Since f'' is continuous on $[a, b]$, there exists a maximum value

$$M = \max_{x \in [a, b]} f''(x) \text{ and a minimum value } m = \min_{x \in [a, b]} f''(x).$$

Thus

$$nm \leq \sum_{i=1}^n f''(\bar{\xi}_i) \leq nM.$$

It then follows by the intermediate value theorem (Appendix A) that there exists a point $\xi \in [a, b]$ such that

$$\sum_{i=1}^n f''(\bar{\xi}_i) = n f''(\xi).$$

Hence we can write the above formula

$$\int_a^b f(x) dx = \frac{h}{2} \left[f(x_0) + f(x_n) + 2 \sum_{i=1}^{n-1} f(x_i) \right] - \frac{h^3}{12} n f''(\xi), \quad \xi \in [a, b]$$

or

$$\int_a^b f(x) dx = \frac{h}{2} \left[f(x_0) + f(x_n) + 2 \sum_{i=1}^{n-1} f(x_i) \right] - \frac{(b-a)}{12} h^2 f''(\xi), \quad \xi \in [a, b]. \quad (11.14)$$

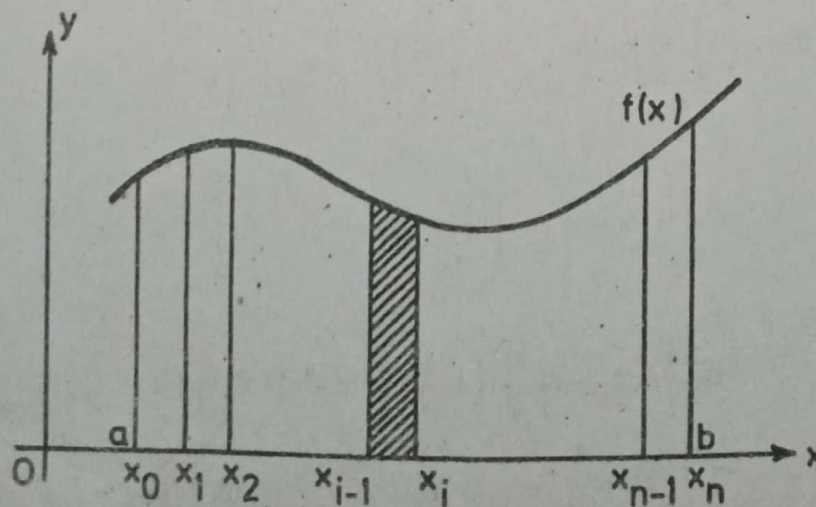


Fig.11.3 Composite Trapezoidal Rule.

In terms of n , the number of application of the rule (the number of subintervals), and a and b , the limits of integration, the composite trapezoidal rule is

$$\int_a^b f(x)dx = \frac{b-a}{2n} \left[f(a) + f(b) + 2 \sum_{i=1}^{n-1} f\left(a + \frac{b-a}{n} i\right) \right] - \frac{(b-a)^3}{12n^2} f''(\xi), \quad \xi \in [a, b]. \quad (11.15)$$

The error for the composite trapezoidal rule is proportional to $1/n^2$. Therefore, if we double the number of applications, the error will decrease roughly by a factor of four [$f''(\xi)$ will usually be different for two different values of n].

11.6.2 COMPOSITE SIMPSON'S RULE

Let $f \in C^4[a, b]$. Subdivide the interval $[a, b]$ into $2m$ equal subintervals. With $h = (b-a)/2m$, and $x_i = a + ih$, $i=0, 1, \dots, 2m$, apply Simpson's rule (11.10) to each sub-interval

$$[x_0, x_2], [x_2, x_4], \dots, [x_{2m-2}, x_{2m}]$$

of length $2h$ (see Fig. 11.4), then

$$\begin{aligned} \int_a^b f(x)dx &= \int_{x_0}^{x_{2m}} f(x)dx \\ &= \int_{x_0}^{x_2} f(x)dx + \int_{x_2}^{x_4} f(x)dx + \dots + \int_{x_{2m-2}}^{x_{2m}} f(x)dx \\ &= \frac{h}{3} \left[f(x_0) + 4f(x_1) + f(x_2) \right] + \frac{h}{3} \left[f(x_2) + 4f(x_3) + f(x_4) \right] \\ &\quad + \dots + \frac{h}{3} \left[f(x_{2m-2}) + 4f(x_{2m-1}) + f(x_{2m}) \right] \\ &\quad - \frac{h^5}{90} \sum_{i=1}^m f^{(4)}(\bar{\xi}_i), \quad \bar{\xi}_i \in (x_{2i-2}, x_{2i}) \text{ for } i=1, 2, \dots, m. \end{aligned}$$

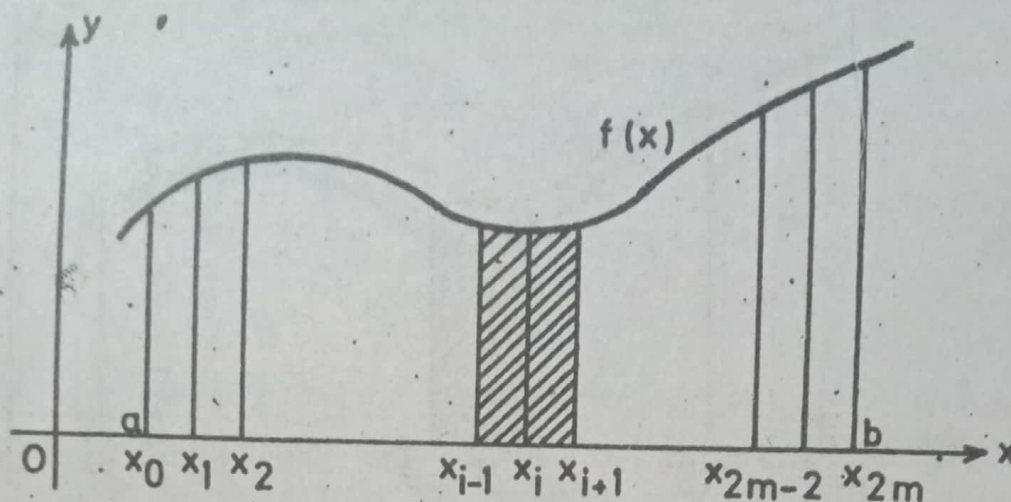


Fig. 11.4 Composite Simpson's Rule

$$\int_a^b f(x) dx = \frac{h}{3} \left[f(x_0) + 4 \left\{ f(x_1) + f(x_3) + \dots + f(x_{2m-1}) \right\} + 2 \left\{ f(x_2) + f(x_4) + \dots + f(x_{2m-2}) \right\} + f(x_{2m}) \right] -$$

$$\frac{h^5}{90} \sum_{i=1}^m f^{(4)}(\bar{\xi}_i), \quad \bar{\xi}_i \in (x_{2i-2}, x_{2i}) \text{ for } i = 1, 2, \dots, m$$

$$= \frac{h}{3} \left[f(x_0) + 4 \sum_{i=1}^m f(x_{2i-1}) + 2 \sum_{i=1}^{m-1} f(x_{2i}) + f(x_{2m}) \right] -$$

$$\frac{h^5}{90} \sum_{i=1}^m f^{(4)}(\bar{\xi}_i), \quad \bar{\xi}_i \in (x_{2i-2}, x_{2i}), \quad i = 1, 2, \dots, m.$$

Since $f \in C^4[a, b]$,

$$\min_{x \in [a, b]} f^{(4)}(x) \leq f^{(4)}(\bar{\xi}_i) \leq \max_{x \in [a, b]} f^{(4)}(x)$$

so

$$m \min_{x \in [a, b]} f^{(4)}(x) \leq \sum_{i=1}^m f^{(4)}(\bar{\xi}_i) \leq m \max_{x \in [a, b]} f^{(4)}(x)$$

or

$$\min_{x \in [a,b]} f^{(4)}(x) \leq \frac{1}{m} \sum_{i=1}^m f^{(4)}(\xi_i) \leq \max_{x \in [a,b]} f^{(4)}(x).$$

By the intermediate value theorem (Appendix A), there exists a point $\xi \in [a,b]$ such that

$$\sum_{i=1}^m f^{(4)}(\xi_i) = m f^{(4)}(\xi).$$

Thus,

$$\int_a^b f(x) dx = \frac{h}{3} \left[f(x_0) + 4 \sum_{i=1}^m f(x_{2i-1}) + 2 \sum_{i=1}^{m-1} f(x_{2i}) + f(x_{2m}) \right] - \frac{h^5}{90} m f^{(4)}(\xi), \quad \xi \in [a,b].$$

Since

$$h = \frac{b-a}{2m}, \quad f(x_0) = f(a), \quad f(x_{2m}) = f(b),$$

$$\begin{aligned} \int_a^b f(x) dx &= \frac{h}{3} \left[f(x_0) + 4 \sum_{i=1}^m f(x_{2i-1}) + 2 \sum_{i=1}^{m-1} f(x_{2i}) + f(x_{2m}) \right] \\ &\quad - \frac{(b-a)}{180} h^4 f^{(4)}(\xi), \quad \xi \in [a,b]. \end{aligned} \quad (11.16)$$

In terms of a , b , and m , (11.16) is given by

$$\begin{aligned} \int_a^b f(x) dx &= \frac{(b-a)}{6m} \left[f(a) + 4 \sum_{i=1}^m f\left(a + \left(\frac{b-a}{2m}\right)(2i-1)\right) + 2 \sum_{i=1}^{m-1} f\left(a + \frac{b-a}{m} i\right) + f(b) \right] \\ &\quad - \frac{(b-a)^5}{2880m^4} f^{(4)}(\xi), \quad \xi \in [a,b]. \end{aligned} \quad (11.17)$$

Composite formulae similar to those of (11.14) and (11.16) can

be generated for any of the low-order integration formulae.

If the maximum admissible error $\epsilon > 0$ is given, then, denoting

$$M_4 = \max |f^{(4)}(x)|,$$

we will have, for determining the spacing h , the inequality

$$(b - a) \frac{h^4}{180} M_4 < \epsilon.$$

Hence $h < \left(\frac{180\epsilon}{(b-a)M_4} \right)^{1/4}$. Thus h is of the order $(\epsilon)^{1/4}$.

EXAMPLE 11.5: Apply the composite Simpson's rule to the integral

$$\int_1^{1.30} \sqrt{x} \, dx \quad \text{taking } 2m = 6.$$

SOLUTION: We have $h = \frac{1.30-1}{6} = .05$.

$$\text{Now, } \int_1^{1.30} \sqrt{x} \, dx = \frac{2}{3} \left[(1.3)^{3/2} - 1 \right] = .32149 \text{ to five decimal places.}$$

Using the composite Simpson's rule, we obtain

$$\begin{aligned} \int_1^{1.3} \sqrt{x} \, dx &= \frac{.05}{3} \left[f(1) + 4 \left\{ f(1.05) + f(1.15) + f(1.25) \right\} + \right. \\ &\quad \left. 2 \left\{ f(1.10) + f(1.20) \right\} + f(1.30) \right] \end{aligned}$$

$$\begin{aligned} \int_1^{1.30} \sqrt{x} \, dx &= \frac{.05}{3} \left[1.0 + 4(1.02470 + 1.07238 + 1.11803) + \right. \\ &\quad \left. 2(1.04881 + 1.09544) + 1.14017 \right] = 0.32149 \end{aligned}$$

which is correct to five decimal places.

EXAMPLE 11.6: Approximate the integral $\int_0^{\pi} \sin x \, dx$ with an error of at most .00002, using the composite Simpson's rule.

SOLUTION: Applying the composite Simpson's rule (11.16) to the given integral we have

$$\int_0^{\pi} \sin x \, dx = \frac{h}{3} \left[\sin 0 + 4 \sum_{i=1}^m \sin(x_{2i-1}) + 2 \sum_{i=1}^{m-1} \sin(x_{2i}) + \sin \pi \right] - \frac{(\pi-0)h^4}{180} \sin \xi$$

$$\int_0^{\pi} \sin x \, dx = \frac{h}{3} \left[4 \sum_{i=1}^m \sin(x_{2i-1}) + 2 \sum_{i=1}^{m-1} \sin(x_{2i}) \right] - \frac{\pi h^4}{180} \sin \xi.$$

Since the error is required to be less than .00002, the inequality

$$\left| \frac{\pi h^4}{180} \sin \xi \right| \leq \frac{\pi h^4}{180} = \frac{\pi \left(\frac{\pi-0}{2m} \right)^4}{180} = \frac{\pi^5}{2880m^4} \leq .00002$$

can be used to determine the value of m and h . From this relation we get $m \geq 9$.

For $m = 10$, $h = \frac{\pi-0}{2m} = \frac{\pi}{20}$, we obtain

$$\int_0^{\pi} \sin x \, dx \approx \frac{\pi}{60} \left[4 \sum_{i=1}^{10} \sin\left(\frac{(2i-1)\pi}{20}\right) + 2 \sum_{i=1}^9 \sin\left(\frac{i\pi}{10}\right) \right] = 2.00000679.$$

To be sure of this degree of accuracy using the composite trapezoidal rule, the truncation error would require that

$$\left| \frac{\pi h^2}{12} \sin \xi \right| \leq \frac{\pi h^2}{12} = \frac{\pi^3}{12n^2} \leq .00002 \text{ or } n \geq 360.$$

Since it involves many calculations, it is clear that it would be undesirable to use the composite trapezoidal rule on this problem. For comparison purposes, the composite trapezoidal rule with $n=20$

$$h = \frac{\pi-0}{20} = \frac{\pi}{20} \text{ gives}$$

$$\begin{aligned} \int_0^{\pi} \sin x \, dx &\approx \frac{\pi}{40} \left[\sin 0 + 2 \sum_{i=1}^{19} \sin\left(\frac{i\pi}{20}\right) + \sin \pi \right] \\ &= \frac{\pi}{40} \left[2 \sum_{i=1}^{19} \sin\left(\frac{i\pi}{20}\right) \right] = 1.9958860. \end{aligned}$$

As
$$\int_0^{\pi} \sin x \, dx = [-\cos x]_0^{\pi} = 2;$$

the composite Simpson's rule gives a value well within the required error bound.

11.7 RICHARDSON'S EXTRAPOLATION

Let I_n and E_n be, respectively, the estimates of the integral and the associated error for n application of the composite trapezoidal rule. Then the true value I of the integral is

$$I = I_{n_1} + E_{n_1} = I_{n_2} + E_{n_2} \quad (11.18)$$

where n_1 and n_2 are two different values of n (the number of sub-intervals) in the composite trapezoidal rule. From (11.15), we obtain

$$\frac{E_{n_1}}{E_{n_2}} = \frac{-(b-a)^3/12n_1^2 f''(\xi_1)}{-(b-a)^3/12n_2^2 f''(\xi_2)}, \quad \xi_1, \xi_2 \in [a, b].$$

Assuming that $f''(\xi_1)$ and $f''(\xi_2)$ are equal, the above equation reduces to

$$\frac{E_{n_1}}{E_{n_2}} = \left(\frac{n_2}{n_1} \right)^2. \quad (11.19)$$

Also, from (11.18), we get

$$E_{n_1} = I - I_{n_1}, \quad E_{n_2} = I - I_{n_2}$$

and consequently,

$$\frac{E_{n_1}}{E_{n_2}} = \frac{I - I_{n_1}}{I - I_{n_2}}. \quad (11.20)$$

Substitution of (11.20) into (11.19) leads to improved value I_e of the integral I :

$$I_e = I_{n_2} + \frac{I_{n_2} - I_{n_1}}{\left(\frac{n_2}{n_1} \right)^2 - 1}. \quad (11.21)$$

When $n_2 = 2n_1$, (11.21) becomes

$$I_e = \frac{4}{3} I_{n_2} - \frac{1}{3} I_{n_1}. \quad (11.22)$$

This kind of approach in which two approximations to an integral are used to get a third better approximation is called **Richardson's extrapolation** or **Richardson's deferred approach to the limit**. Richardson's extrapolation may be applied to any numerical process in which the truncation error is (approximately) proportional to some power of the step size.

In general, if an integration formula being used has a truncation error of order h^m , then

$$I_e = I_{n_2} + \frac{I_{n_2} - I_{n_1}}{\left(\frac{h_1}{h_2} \right)^{m-1} - 1} \quad (11.23)$$

where I_e is the extrapolated value of the integral and I_{n_1} , I_{n_2} respectively, denote the estimates of the integral obtained by using two different step sizes h_1 and h_2 .

If n_1 , n_2 respectively, are the number of subintervals used in the quadrature formula, then (11.23) becomes

$$I_e = I_{n_2} + \frac{I_{n_2} - I_{n_1}}{\left(\frac{n_2}{n_1}\right)^{m-1} - 1} \quad (11.24)$$

In particular, if $h_2 = h_1/2$, that is, $n_2 = 2n_1$, then

$$I_e = I_{n_2} + \frac{I_{n_2} - I_{n_1}}{2^{m-1} - 1} \quad (11.25)$$

For the commonly used integration formulae, Richardson's extrapolation takes the following forms:

Trapezoidal rule ($m = 3$):

$$I_e = I_{n_2} + \frac{I_{n_2} - I_{n_1}}{3} = \frac{4}{3} I_{n_2} - \frac{1}{3} I_{n_1}$$

Simpson's 1/3 & 3/8 rules ($m = 5$):

$$I_e = I_{n_2} + \frac{I_{n_2} - I_{n_1}}{15} = \frac{16}{15} I_{n_2} - \frac{1}{15} I_{n_1}$$

EXAMPLE 11.7: Evaluate the integral

$$I = \int_1^3 [x^3 - 2x^2 + 7x - 5] dx$$

by using first two and then four applications of the composite trapezoidal rule. Use Richardson's extrapolation to find a third estimate of the integral.

SOLUTION: $I = \int_1^3 (x^3 - 2x^2 + 7x - 5) dx = 20 \frac{2}{3}$.

Let $n_1 = 2$ and $n_2 = 4$. Then Eq. (11.14) yields.

$$I_{n_1} = \frac{1}{2} [f(1) + f(3) + 2f(2)] = \frac{1}{2} [1 + 25 + 2 \times 9] = 22$$

$$I_{n_2} = \frac{1}{2 \times 2} [f(1) + f(3) + 2f(1.5) + 2f(2) + 2f(2.5)]$$

$$= \frac{1}{4} [1 + 25 + 2 \times 4 \frac{3}{8} + 2 \times 9 + 2 \times 15 \frac{5}{8}] = 21.$$

Since $n_2 = 2n_1$, substitution of I_{n_1} , I_{n_2} into Eq. (11.22) gives a third estimate,

$$I_e = \frac{4}{3}(21) - \frac{1}{3}(22) = 20 \frac{2}{3} = I.$$

EXERCISE 11.2

- Determine the values of m and h needed to approximate $\int_1^3 e^x dx$, with error less than 10^{-6} , using composite Simpson's rule.
- Approximate $\int_1^3 \ln x dx$ to within 10^{-2} , using composite Simpson's rule.
- Estimate the integral $\int_0^1 e^{\sin x} dx$ using the composite trapezoidal rule with $h=1$, $1/2$ and $1/4$. Improve the estimates by Richardson's extrapolation.
- Use the composite trapezoidal rule to estimate $\int_{e-1}^5 f(x) dx$ for the following tabulated function:

$u_0 \quad u_1 = u_0 + h \quad u_2 = u_1 + h$

x	0.1	0.2	0.3	0.4	0.5
$f(x)$	0.9950	0.9801	0.9553	0.9211	0.8776

Use Simpson's rule to approximate the same integral.

5. Use the composite trapezoidal rule to approximate an integral $\int_0^{1.5} (1+x)^{-1} dx$ taking $n = 2$.
6. Repeat exercise 5, using the composite Simpson's rule with $m = 5$.
7. Let $f(x)$ be defined by

$$f(x) = \begin{cases} x^3 + 1, & 0 \leq x \leq .1 \\ 1.001 + .03(x-.1) + .3(x-.1)^2 + 2(x-.1)^3, & .1 \leq x \leq .2 \\ 1.009 + .15(x-.2) + .9(x-.2)^2 + 2(x-.2)^3, & .2 \leq x \leq .3. \end{cases}$$

- a) Approximate $\int_0^3 f(x)dx$, using the extended (composite)

trapezoidal rule with $n = 6$.

- b) Approximate $\int_0^3 f(x)dx$, using Simpson's composite rule with $m = 6$. Are the results more accurate than in part (a)?

8. Let $f(x)$ be defined by

$$f(x) = \begin{cases} x^2 - 1, & 0 \leq x \leq .5 \\ e^x \sin x, & .5 < x \leq 1. \end{cases}$$

- a) Evaluate $\int_0^1 f(x)dx$.

- b) Approximate the integral in (a) using Simpson's composite rule with $h = .1$.

- a 9. Apply the composite trapezoidal rule to approximate the integral $\int_{1.0}^{1.3} \sqrt{x} dx$ with $h = .05$. Compare with correct value of the integral.

10. Apply the composite Simpson's rule to the integral of exercise 9.

$$h = \frac{b-a}{n}$$

$$0.05 = \frac{1.3 - 1.0}{n} \Rightarrow n = 6$$

11. Estimate the truncation error in exercise 9.
12. Estimate the truncation error in exercise 10.
13. Use the truncation error estimate for the trapezoidal rule to predict how tightly values of $f(x)$ must be packed (what interval h) for the trapezoidal rule itself to achieve a correct result to four decimal places for $\int_1^2 \frac{1}{x} dx$.
14. Use the truncation error estimate for Simpson's rule to predict how many values of $f(x)$, or how small an interval h , will be needed for this rule to produce $\ln 2$ correct to four decimal places.
15. How small an interval h would be required to obtain $\ln 2$ correct to eight decimal places using the trapezoidal rule? Using Simpson's rule?
16. Apply Richardson's extrapolation to the evaluation of

$$\int_{0.2}^{1.0} e^{-x} dx$$

from the data

x	0.2	0.4	0.6	0.8	1.0
f(x)	0.818731	0.670320	0.548812	0.449329	0.367879

using the trapezoidal rule and Simpson's 1/3 rule. If the exact value is 0.450851 (to six significant figures) then compute the actual errors in different integration formulae.

11.8 NEWTON-COTES CLOSED QUADRATURE FORMULAE

Suppose, for a given function $y=f(x)$, it is required to compute the integral

$$\int_a^b f(x) dx.$$

Choosing a spacing $h = \frac{b-a}{n}$

divide the given interval $[a,b]$ into n equal parts by means of equally spaced points

$$x_i = x_0 + ih, \quad i = 0, 1, 2, \dots, n.$$

For Newton-Cotes closed formula, we have $x_0 = a$, $x_n = b$.

Let $y_i = f(x_i), \quad i = 0, 1, \dots, n.$

Replacing the function $f(x)$ by an appropriate Lagrange interpolation polynomial $P_n(x)$, we get the approximate quadrature formula

$$\int_{x_0}^x f(x) dx = \sum_{i=0}^n a_i f(x_i)$$

where a_i are certain constant coefficients given by

$$a_i = \int_a^b L_i(x) dx, \quad i = 0, 1, \dots, n.$$

$$a_i = \int_{x_0}^x \frac{(x-x_0)(x-x_1)\dots(x-x_{i-1})(x-x_{i+1})\dots(x-x_n)}{(x_1-x_0)(x_1-x_1)\dots(x_1-x_{i-1})(x_1-x_{i+1})\dots(x_1-x_n)} dx$$

Substituting $x = x_0 + th$, then

$$x-x_j = x_0 + th - (x_0 + jh) = h(t-j), \quad x_i - x_j = x_0 + ih - (x_0 + jh) = h(i-j),$$

and consequently we get

$$\begin{aligned} a_i &= h \int_0^n \frac{t(t-1)(t-2)\dots(t-i-1)(t-i+1)\dots(t-n)}{i(i-1)(i-2)\dots 1 \cdot (-1)(-2)\dots(i-n)} dt \\ &= h \int_0^n \frac{t(t-1)\dots(t-i-1)(t-i+1)\dots(t-n)}{i!(-1)(-2)\dots(-(n-i))} dt \end{aligned}$$

$$a_i = h \frac{(-1)^{n-i}}{i!(n-i)!} \int_0^n t(t-1)\dots(t-i-1)(t-i+1)\dots(t-n) dt,$$

$$i = 0, 1, \dots, n. \quad (11.26)$$

Thus the quadrature formula is

$$\int_a^b f(x)dx = \sum_{i=0}^n a_i f(x_i) + E_{n+1}(f), \quad (11.27)$$

where a_i 's are defined by (11.26) and the error term (see Theorem 11.3) is

$$E_{n+1}(f) = \begin{cases} \frac{h^{n+2} f^{(n+1)}(\xi)}{(n+1)!} \int_0^n t(t-1)\dots(t-n)dt & \text{if } n \text{ is odd} \\ \frac{h^{n+3} f^{(n+2)}(\xi)}{(n+2)!} \int_0^n t^2(t-1)\dots(t-n)dt & \text{if } n \text{ is even} \end{cases} \quad (11.28)$$

and $f \in C^{n+1}[a, b]$,
and $f \in C^{n+2}[a, b]$.

Some of the Newton-Cotes closed quadrature formulae together with their respective truncation errors are listed below.

$n = 1$ (Trapezoidal Rule)

$$\int_{x_0}^{x_1} f(x)dx = \frac{h}{2} [f(x_0) + f(x_1)] - \frac{h^3}{12} f^{(2)}(\xi), \quad \xi \in (x_0, x_1) \quad (11.29)$$

$n = 2$ (Simpson's First or 1/3 Rule):

$$\int_{x_0}^{x_2} f(x)dx = \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)] - \frac{h^5}{90} f^{(4)}(\xi), \quad \xi \in (x_0, x_2) \quad (11.30)$$

$n = 3$ (Simpson's Second or 3/8 Rule):

$$\int_{x_0}^{x_3} f(x)dx = \frac{3}{8} h [f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)] - \frac{3h^5}{80} f^{(4)}(\xi), \quad \xi \in (x_0, x_3). \quad (4.31)$$

$n = 4$ (Boole's Rule):

$$\int_{x_0}^{x_4} f(x)dx = \frac{2h}{45} [7f(x_0) + 32f(x_1) + 12f(x_2) + 32f(x_3) + 7f(x_4)] - \frac{8h^7}{945} f^{(6)}(\xi), \quad \xi \in (x_0, x_4). \quad (11.32)$$

$n = 5$:

$$\int_{x_0}^{x_5} f(x) dx = \frac{5h}{288} \left[19 \left\{ f(x_0) + f(x_5) \right\} + 75 \left\{ f(x_1) + f(x_4) \right\} + 50 \left\{ f(x_2) + f(x_3) \right\} \right] - \frac{275h^7}{12096} f^{(6)}(\xi), \quad \xi \in (x_0, x_5). \quad (11.33)$$

$n = 6$:

$$\int_{x_0}^{x_6} f(x) dx = \frac{h}{140} \left[41 \left\{ f(x_0) + f(x_6) \right\} + 216 \left\{ f(x_1) + f(x_5) \right\} + 27 \left\{ f(x_2) + f(x_4) \right\} + 272 f(x_3) \right] - \frac{9h^9}{1400} f^{(8)}(\xi), \quad \xi \in (x_0, x_6). \quad (11.34)$$

There is another seven-point formula known as Weddle's rule. In this rule the co-efficients of $f(x_i)$, ($i = 0, 1, \dots, 6$) are very simple. This rule is

$$\int_{x_0}^{x_6} f(x) dx = \frac{3h}{10} \left[f(x_0) + 5f(x_1) + f(x_2) + 6f(x_3) + f(x_4) + 5f(x_5) + f(x_6) \right] - \frac{h^7}{140} f^{(6)}(\xi), \quad \xi \in (x_0, x_6).$$

Like trapezoidal's rule and Simpson's rule, Weddle's rule can be extended to an interval of integration by dividing it into subintervals as a multiple of six. In the case of Boole's rule the interval is divided into subintervals as a multiple of four.

The geometric meaning of Weddle's rule is that we replace the graph of the given function by $n/6$ arcs, n being the number of subintervals, of a fifth-degree polynomial.

Note that when n is even (that is, when there is an even number of subintervals or an odd number of points) the formulae are exact for a polynomial of degree $n+1$ or less; when n is odd, the formulae are exact for a polynomial of degree n or less. For this reason the odd-point formulae are used more frequently than the even-point formulae.

If $f(x)$ is known analytically, then it may be possible to determine the appropriate high-order derivative and examine its behavior on the interval $[a, b]$ to establish an error bound for the particular formula chosen. In many practical situations, however, the error formula is of little direct value since an expres-

ssion for the derivative may not be available.

Even when one has no information about higher order derivatives, it may be possible to estimate the error if the integral is computed using two different integration formulae with comparable degree of precision. For example, consider the evaluation of the integral

$$I = \int_a^b f(x) dx$$

by using Simpson's 1/3 and 3/8 rules, each of degree of precision three. Let I_1 and E_1 be the estimate of I and the error, respectively, resulting from the use of Simpson's 1/3 rule, and let I_2 and E_2 be like quantities, resulting from the use of Simpson's 3/8 rule. Then

$$I = I_1 + E_1 = I_2 + E_2$$

or

$$E_1 = I - I_1 \text{ and } E_2 = I - I_2$$

which gives

$$\frac{E_1}{E_2} = \frac{I - I_1}{I - I_2} \quad (11.35)$$

In terms of the limits of integration a and b , we have

$$\frac{E_1}{E_2} = \frac{-\frac{(b-a)^5}{2880} f^{(4)}(\xi_1)}{-\frac{(b-a)^5}{6480} f^{(4)}(\xi_2)}, \quad \xi_1, \xi_2 \in (a, b).$$

If we assume that $f^{(4)}(\xi_1)$ and $f^{(4)}(\xi_2)$ are approximately equal, then the above equation reduces to

$$\frac{E_1}{E_2} = \frac{9}{4} \quad (11.36)$$

Using (11.36) and by virtue of (11.35), we get

$$I_e = \frac{9}{5} I_2 - \frac{4}{5} I_1 \quad (11.37)$$

EXAMPLE 11.8: Evaluate

$$I = \int_0^1 \frac{dx}{1+x}$$

using a Newton-Cotes closed formula employing seven ordinates ($n = 6$).

SOLUTION: $I = \int_0^1 \frac{1}{1+x} dx = \left[\ln(1+x) \right]_0^1 = \ln 2 = .69315.$

Let us, now, approximate the integral using a Newton-Cotes closed formula for $n = 6$. Take $h = (1-0)/6 = 1/6$. Then

$$\begin{aligned} I &= \int_0^1 \frac{1}{1+x} dx \approx \frac{1/6}{140} \left[41 \left\{ f(0) + f(1) \right\} + 216 \left\{ f\left(\frac{1}{6}\right) + f\left(\frac{5}{6}\right) \right\} \right. \\ &\quad \left. + 27 \left\{ f\left(\frac{1}{3}\right) + f\left(\frac{2}{3}\right) \right\} + 272 f\left(\frac{1}{2}\right) \right] \\ I &\approx \frac{1}{740} \left[41 \left\{ 1 + \frac{1}{2} \right\} + 216 \left\{ \frac{6}{7} + \frac{6}{11} \right\} + 27 \left\{ \frac{3}{4} + \frac{3}{5} \right\} + 272 \times \frac{2}{3} \right] \\ &\approx \frac{1}{740} \left[41 \times \frac{3}{2} + 216 \times \frac{108}{77} + 27 \times \frac{27}{20} + \frac{544}{3} \right] = 0.6933. \end{aligned}$$

EXERCISE 11.3

1. Use the Newton-Cotes closed formula for $n = 3$ to $\int_1^3 e^{-x/2} dx$. Compare approximation obtained with actual value 0.7668010.

2. Using the composite Simpson's rule, compute the integral

$$I = \int_0^1 \frac{dx}{1+x}$$

taking $m = 5$.

3. Approximate the integrals

$$\text{a) } \int_0^{.1} \sqrt{1+x} \, dx \quad \text{b) } \int_{1.1}^{1.5} e^x \, dx$$

using the Newton-Cotes closed formulae for $n = 2, 3, 4$.

4. Approximate the integral $\int_1^{1.5} e^{-x^2} \, dx$, using Newton-Cotes closed formulae with $n = 1, 2, 3, 4$.

11.9 THE METHOD OF UNDETERMINED CO-EFFICIENTS

The integration formulae we have discussed so far are of the type

$$\int_a^b f(x) \, dx \approx A_1 f(x_1) + A_2 f(x_2) + \dots + A_n f(x_n) \quad (11.38)$$

For example the trapezoidal rule has $n=2$, $a=x_0$, $b=x_0+h$, $x_1=x_0$, $x_2=x_0+h$, $A_1=A_2=\frac{h}{2}$. Simpson's $\frac{1}{3}$ formula has $n=3$, $a=x_0$, $b=x_0+2h$, $x_1=x_0$, $x_2=x_0+h$, $x_3=x_0+2h$, $A_1=\frac{h}{3}$, $A_2=\frac{4h}{3}$, and $A_3=\frac{h}{3}$. We can obtain any number of new integration formulae by assuming an expression of the form (11.38) and choosing A_i ($i=1, \dots, n$) in a manner so as to make the error as small as possible. If we choose $x_1, x_2, \dots, x_n$ in some manner, say equally spaced points with $a=x_1$ and $b=x_n$, then we have n parameters $A_1, A_2, \dots, A_n$ at our disposal. We can choose them by requiring that (11.38) should be exact for every polynomial of degree less than or equal to $n-1$. Equivalently we can require that (11.38) should be exact, when we successively substitute $f(x) = 1, x, x^2, \dots, x^{n-1}$.

EXAMPLE 11.9: Let $n = 2$, $x_1 = a$, $x_2 = a + h$, then (11.38) becomes

$$\int_a^{a+h} f(x) \, dx \approx A_1 f(a) + A_2 f(a+h). \quad (11.39)$$

Let $f(x) = 1$, then (11.39) becomes

$$h = A_1 + A_2 \quad (11.40)$$

Let $f(x) = x$ and we get

$$\frac{(a+h)^2}{2} - \frac{a^2}{2} = A_1 a + A_2 (a+h)$$

$$ha + \frac{h^2}{2} = a(A_1 + A_2) + A_2 h.$$

If we make use of (11.40), the above equation reduces to

$$\frac{h^2}{2} = A_2 h. \quad (11.41)$$

Now we can solve (11.40) and (11.41) and find

$$A_1 = A_2 = \frac{h}{2}.$$

Thus we have re-derived the trapezoidal rule.

In the next example we choose the origin as the mid point of the interval of integration. This can always be achieved by a transformation of co-ordinates.

EXAMPLE 11.10: Let $\int_{-h}^h f(x) dx \approx A_1 f(-h) + A_2 f(0) + A_3 f(h).$ (11.42)

We require that (11.42) should be exact for all polynomials of degree upto 2. By taking successively $f(x) = 1$, x and x^2 , we find

$$f(x) = 1: \quad 2h = A_1 + A_2 + A_3 \quad (11.43)$$

$$f(x) = x: \quad 0 = -hA_1 + hA_3 \quad (11.44)$$

$$f(x) = x^2: \quad \frac{2h^3}{3} = h^2 A_1 + h^2 A_3 \quad (11.45)$$

We can solve the above system and find

$$A_1 = A_3 = \frac{h}{3}, \quad A_2 = \frac{4}{3} h.$$

We recognize that we have obtained Simpson's $\frac{1}{3}$ formula. We note that, although we required (11.42) to be exact for polynomials upto degree 2, the formula actually gives exact result even for

every polynomial of degree 3. This can be seen by putting $f(x)=x^3$ in (11.42). We get

$$0 = -h^3 A_1 + h^3 A_3$$

or

$$0 = A_1 - A_3$$

which is the same equation as (11.44). This explains the fact why a Newton-Cotes formula with $2k$ panels is as accurate or even more accurate as compared with a formula with $2k+1$ panels.

11.10 GAUSSIAN QUADRATURE

Before we discuss this topic we briefly state a few results concerning Legendre polynomials. For details the reader is referred to [10]. The Legendre polynomial $P_n(x)$ is a polynomial of degree n which can be obtained from the formula

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n.$$

First few polynomials are:

$$\left. \begin{aligned} P_0(x) &= 1, \\ P_1(x) &= x, \\ P_2(x) &= \frac{3x^2 - 1}{2}, \\ P_3(x) &= \frac{5x^3 - 3x}{2}, \\ P_4(x) &= \frac{35x^4 - 30x^2 + 3}{8} \end{aligned} \right\} \quad (11.46)$$

The set $\{P_n(x)\}_{n=0}^{\infty}$ is orthogonal i.e.

$$\int_{-1}^1 P_m(x) P_n(x) dx = \begin{cases} 0 & \text{if } m \neq n \\ \frac{2}{2n+1} & \text{if } m = n. \end{cases} \quad (11.47)$$

Every polynomial of degree k can be written as a linear combination of $P_0, P_1(x), \dots, P_k(x)$. For example from (11.46) we have

$$x = P_1(x) \quad (11.48)$$

$$\begin{aligned} 3x^2 &= 2P_2(x) + 1, \\ &= 2P_2(x) + P_0(x), \end{aligned}$$

$$\text{or} \quad x^2 = \frac{2}{3} P_2(x) + \frac{1}{3} P_0(x). \quad (11.49)$$

$$\begin{aligned} \text{Also} \quad 5x^3 &= 2P_3(x) + 3x, \\ &= 2P_3(x) + 3P_1(x), \end{aligned}$$

$$\text{or} \quad x^3 = \frac{2}{5} P_3(x) + \frac{3}{5} P_1(x). \quad (11.50)$$

Since x^k can be expressed as a linear combination of $P_0, P_1, \dots, P_k$ we can use the orthogonality relation (11.47) to show that

$$\int_{-1}^1 x^k P_n(x) dx = 0 \quad (k = 0, 1, 2, \dots, n-1). \quad (11.51)$$

Finally we mention that the equation $P_n(x) = 0$ has n roots which are real, distinct and all of them lie in the interval $(-1, 1)$.

A Gaussian quadrature formula is of the type

$$\int_{-1}^1 f(x) dx \approx A_1 f(x_1) + A_2 f(x_2) + \dots + A_n f(x_n) \quad (11.52)$$

where not only A_i ($i = 1, \dots, n$) but also $x_1, x_2, \dots, x_n$ are to be chosen in such a way that the formula is as best as possible. Since we have $2n$ parameters at our disposal we require that the formula should be exact for all polynomials of degree upto $2n-1$. Proceeding as in the last section and successively taking

$$f(x) = 1, x, x^2, \dots, x^{2n-1}$$

we obtain

$$\begin{aligned}
 f(x) &= 1, & 2 &= A_1 + A_2 + \dots + A_n \\
 f(x) &= x, & 0 &= A_1 x_1 + A_2 x_2 + \dots + A_n x_n \\
 f(x) &= x^2, & \frac{2}{3} &= A_1 x_1^2 + A_2 x_2^2 + \dots + A_n x_n^2 \\
 & \vdots & & \\
 f(x) &= x^{n-1}, & \frac{2}{n} [1 - (-1)^n] &= A_1 x_1^{n-1} + A_2 x_2^{n-1} + \dots + A_n x_n^{n-1} \\
 f(x) &= x^n, & \frac{2}{n+1} [1 - (-1)^{n+1}] &= A_1 x_1^n + A_2 x_2^n + \dots + A_n x_n^n \\
 & \vdots & & \\
 f(x) &= x^{2n-1}, & 0 &= A_1 x_1^{2n-1} + A_2 x_2^{2n-1} + \dots + A_n x_n^{2n-1}
 \end{aligned} \quad (11.53)$$

We have a system of $2n$ equations in $2n$ unknowns i.e. $A_1, A_2, \dots, A_n, x_1, x_2, \dots, x_n$. However the system is nonlinear and very difficult to solve. To overcome this problem we proceed as follows:

Instead of demanding that (11.52) should be exact for $f(x)=1, x, x^2, \dots, x^{2n-1}$, we require that this should give exact result for each of $f(x) = 1, x, \dots, x^{n-1}, P_n(x), xP_n(x), \dots, x^{n-1}P_n(x)$. Since $P_n(x)$ is a polynomial of degree n and (11.52) is exact for $1, x, \dots, x^{n-1}$ so it will be exact for $P_n(x)$ only if it is exact for x^n , hence the modified requirement will also lead us in the right direction. Now we have a new system of equations whose first n members are the same as in (11.53). The next n members are replaced by

$$\begin{aligned}
 \int_{-1}^1 P_n(x) dx &= A_1 P_n(x_1) + A_2 P_n(x_2) + \dots + A_n P_n(x_n) \\
 \int_{-1}^1 x P_n(x) dx &= A_1 x_1 P_n(x_1) + A_2 x_2 P_n(x_2) + \dots + A_n x_n P_n(x_n) \\
 & \vdots \\
 \int_{-1}^1 x^{n-1} P_n(x) dx &= A_1 x_1^{n-1} P_n(x_1) + A_2 x_2^{n-1} P_n(x_2) + \dots + A_n x_n^{n-1} P_n(x_n)
 \end{aligned} \quad (11.54)$$

Now (11.51) tells us that all the left hand members of the system (11.54) vanish. Then all the equations are satisfied if we choose $x_1, x_2, \dots, x_n$ as the roots of the equation

$$P_n(x) = 0. \quad (11.55)$$

With $x_i (i = 1, 2, \dots, n)$ fixed as the roots of (11.55) substitute for these in the first n equations of (11.53) which are linear in $A_i (i = 1, 2, \dots, n)$ and can be easily solved.

Let us take $n = 3$, then (11.52) becomes

$$\int_{-1}^1 f(x) dx = A_1 f(x_1) + A_2 f(x_2) + A_3 f(x_3).$$

If this formula is to be exact for $f(x) = 1, x, x^2, P_3(x), xP_3(x), x^2P_3(x)$, we must have

$$\left. \begin{aligned} f(x) = 1, & \quad 2 = A_1 + A_2 + A_3, \\ f(x) = x, & \quad 0 = A_1 x_1 + A_2 x_2 + A_3 x_3, \\ f(x) = x^2, & \quad \frac{2}{3} = A_1 x_1^2 + A_2 x_2^2 + A_3 x_3^2, \\ f(x) = P_3(x), & \quad 0 = A_1 P_3(x_1) + A_2 P_3(x_2) + A_3 P_3(x_3), \\ f(x) = xP_3(x), & \quad 0 = A_1 x_1 P_3(x_1) + A_2 x_2 P_3(x_2) + A_3 x_3 P_3(x_3), \\ f(x) = x^2 P_3(x), & \quad 0 = A_1 x_1^2 P_3(x_1) + A_2 x_2^2 P_3(x_2) + A_3 x_3^2 P_3(x_3). \end{aligned} \right\} (11.56)$$

Last three equations are satisfied if we choose x_1, x_2, x_3 which are the roots of $P_3(x) = 0$, i.e.

$$5x^3 - 3x = 0.$$

Thus we can take $x_1 = -\sqrt{0.6}, x_2 = 0, x_3 = \sqrt{0.6}$. When we substitute these values in the first three equations of (11.56), we get

$$\left. \begin{aligned} A_1 + A_2 + A_3 &= 2 \\ A_1 - A_3 &= 0 \\ A_1 + A_3 &= 10/9 \end{aligned} \right\} \quad (11.57)$$

The system (11.57) has the solution $A_1 = 5/9$, $A_2 = 8/9$, $A_3 = 5/9$. We have now a Gaussian quadrature formula with three points

$$\int_{-1}^1 f(x) dx \approx \frac{5}{9} f(-\sqrt{0.6}) + \frac{8}{9} f(0) + \frac{5}{9} f(\sqrt{0.6}). \quad (11.58)$$

Let us apply formula (11.58) to calculate $\int_{-1}^1 \ln(3+t) dt$. We find

$$\begin{aligned} \int_{-1}^1 \ln(3+t) dt &= \frac{5}{9} \ln(3 - \sqrt{0.6}) + \frac{8}{9} \ln 3 + \frac{5}{9} \ln(3 + \sqrt{0.6}) \\ &= 0.44441 + 0.97654 + 0.73794 \\ &= 2.15889. \end{aligned}$$

The exact result is $\ln(64) - 2 = 2.158883$. We find that the approximate result agrees with the exact one to 4 figures after the decimal point.

In the above treatment we have taken $n = 3$. The cases for other values are dealt with in a similar manner. We summarize the results upto $n = 4$ in the following table.

n	x_k	A_k
2	$-\frac{1}{\sqrt{3}}$	1
	$\frac{1}{\sqrt{3}}$	1
3	$-\sqrt{\frac{3}{5}}$	$\frac{5}{9}$
	0	$\frac{8}{9}$
	$\sqrt{\frac{3}{5}}$	$\frac{5}{9}$
4	-0.861137	0.34785
	-0.33998	0.65215
	0.33998	0.65215
	0.861137	0.34785.

If we have an integral with arbitrary limits of integration we can convert it into the standard form i.e. with limits -1 and 1, by a co-ordinate transformation. To see this consider the following integral

$$\int_a^b f(x) dx.$$

Let

$$x = \frac{b-a}{2} t + \frac{b+a}{2}. \quad (11.59)$$

It is easily seen that when $x = a$, $t = -1$ and when $x = b$, $t = 1$. Thus

$$\int_a^b f(x) dx = \frac{b-a}{2} \int_{-1}^1 f\left(\frac{b-a}{2} t + \frac{b+a}{2}\right) dt. \quad (11.60)$$

We can apply the Gaussian quadrature formula to the integral on the right hand side of the above equation.

EXAMPLE 11.11: Evaluate $\int_0^1 \sin(x^2 + 2) dx$.

SOLUTION: We use the transformation $x = \frac{1}{2} t + \frac{1}{2}$. The given integral transforms into

$$\frac{1}{2} \int_{-1}^1 \sin\left(\frac{1}{4} (t+1)^2 + 2\right) dt.$$

If we use 3 point formula we get

$$\int_0^1 \sin(x^2+2) dx = \frac{5}{9} g(-\sqrt{0.6}) + \frac{8}{9} g(0) + \frac{5}{9} g(\sqrt{0.6}),$$

where $g(x) = \frac{1}{2} \sin\left(\frac{1}{4} (t+1)^2 + 2\right)$.

An approximate value of the integral is found to be 0.770368.

EXERCISE 11.4

1. Assume an integration formula of the form

$$\int_0^{2h} f(x) dx \approx A_1 f(-h) + A_2 f(0) + A_3 f(h).$$

Determine A_1 , A_2 , A_3 , so that the above formula should be exact for all polynomials of degree upto 2.

2. Assume a formula for the second derivative of $f(x)$.

$$f''(0) = A_1 f(-h) + A_2 f(0) + A_3 f(h).$$

Determine A_1 , A_2 , and A_3 so that the formula is as good as possible.

3. Evaluate $\int_{-1}^1 \frac{dx}{\sqrt{2+\sin^2 x}}$ by using Gaussian quadrature formula with 4 points.

4. Evaluate $\int_2^3 e^{-2x} \sin \frac{x}{2} dx$ by using Gaussian quadrature formula with three points. Compare with the exact result and find relative error.

5. Laguerre polynomials are defined

$$L_n(x) = e^x \frac{d^n}{dx^n} (e^{-x} x^n), \quad n = 0, 1, 2, \dots$$

They are orthogonal on $(0, \infty)$ with respect to e^{-x} i.e.

$$\int_0^{\infty} L_n(x) L_m(x) e^{-x} dx = 0, \quad \text{if } n \neq m.$$

Write down first five Laguerre polynomials and develop a

theory for the approximate evaluation of integrals of the form

$$\int_0^{\infty} e^{-x} f(x) dx.$$